GEK-83898

Instructions

FUJITSU DISK DRIVE

OPERATION AND MAINTENANCE MANUAL

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WARNING

THIS EQUIPMENT CONTAINS A POTENTIAL HAZARD OF ELECTRICAL SHOCK OR BURN. ONLY PERSONNEL WHO ARE ADEQUATELY TRAINED AND THOROUGHLY FAMILIAR WITH THE EQUIPMENT AND THESE INSTRUCTIONS SHOULD INSTALL, OPERATE OR MAINTAIN THIS EQUIPMENT.

ISOLATION OF TEST EQUIPMENT FROM THE EQUIPMENT UNDER TEST PRESENTS POTENTIAL ELECTRICAL HAZARDS. IF THE TEST EQUIPMENT CANNOT BE GROUNDED TO THE EQUIPMENT UNDER TEST, THE TEST EQUIPMENT'S CASE MUST BE SHIELDED TO PREVENT CONTACT BY PERSONNEL.

TO MINIMIZE HAZARD OF ELECTRICAL SHOCK OR BURN, APPROVED GROUNDING PRACTICES AND PROCEDURES MUST BE FOLLOWED.

CAUTION

THIS EQUIPMENT IS DESIGNED TO WORK IN A NORMAL INDUSTRIAL ATMOSPHERE, OPERATING RADIO TRANSMITTERS NEARBY MAY CAUSE THE CONTROL EQUIPMENT TO MALFUNCTION. IT IS RECOMMENDED THAT RADIO TRANSMITTERS NOT BE OPERATED WITHIN THE VICINITY OF THE CONTROL EQUIPMENT WHEN THE PANEL DOORS ARE OPEN, NOR AT ANY TIME WITHIN THE CONFINES OF A POWER CONTROL ROOM.

Safety Symbol Legend

WARNING Commands attention to an operating procedure, practice, condition or statement, which, if not strictly observed, could result in personal injury or death.

CAUTION Commands attention to an operating procedure, practice, condition or statement, which, if not strictly observed, could result in damage to or destruction of equipment.

NOTE Commands attention to an essential operating or maintenance procedure, condition, or statement that must be highlighted.

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GENERAL

This manual describes the operation and maintenance of the 84 megabyte Fujitsu 8-inch rigid disk drive system, consisting of model M2312 disk drive with control panel and model M231XK power supply.

Using Winchester type technology the disk drive contains non-removable disks in a sealed module with a closed loop air filtration system. The unit has contact start/stop heads that are returned to a landing zone out of the data area during spindle start and stop functions. A rotary actuator using a closed loop servo performs head positioning.

The heads, with an LSI amplifier mounted on each arm, disk, and positioning mechanism are in the sealed module which is not field serviceable.

The drive's serviceable electronics are located on three easily changeable Printed Circuit Boards (PCB'S). Two are mounted on the topside and one on the backside of the drive.

Elastic shock insulators are used in mounting the disk drive in its mounting frame such that the unit does not touch any other hard parts such as mounting plate, etc, when operating. This must also be maintained when non-operating (both storage and shipping).

The M231XK power supply provides the following voltages to the drive:

DC Voltage	Load
+5V ±5%	3.5A
-12V ±5%	3.0A
+ 24V ± 10%	3.0A (typical) 6.0A (maximum)

The +5V DC and -12V DC load is constant during operation but the +24V DC load current will vary during a power up sequence, DC motor accelaration, and/or seek operation.

DISK DRIVE

SPECIFICATIONS

Weight	approx. 27 LBS

DATA RECORDING

ITEM	SPECIFICATION
Storage Capacity (Unformatted)	84,439,040 Bytes
Number of Cylinders	589
Tracks per Cylinder	7
Cylinder Capacity	143,360 Bytes
Track Capacity	20,480 Bytes
Average Rotational Latency	8.3 ms
Rotational Speed	3600 RPM ± 1%
Transfer Rate	1.229MB/sec
Encoding Method	MFM (Modified Freq. Modulation)
Interface Data	NRZ (Non-Return-to-Zero)
Recording Density	9,550 Bits/in (BPI)
Track Density	720 Tracks/in (TPI)
Start/Stop Time	<40 sec
Interface	SMD
Number of Sectors	128 (Maximum)

ENVIRONMENTAL SPECIFICATIONS

OPERATING

Temperature	41" F to 104" F (5" C to 40" C)
Relative Humidity	20% to 80% (non condensing}
Temperature Variation	<27° F/hr (15" C/hr)
Altitude	10,000 feet
Vibration	0.2G max. (3 Hz to 60 Hz)
Shock	< 2.0G, 10ms

NON-OPERATING

Temperature	-40" F to 140" F (-40" C to 60° C)
Relative Humidity	5% to 95% (non condensing)
Altitude	40,000 feet
Vibration	0.4G max. (3 Hz to 60 Hz)
Shock	< 5.0G, 30ms

OPERATION

This secton contains the operating instructions along with a description of the operator controls and indicators for the M2312 Disk drive.

LOCKING/UNLOCKING THE ACTUATOR

The disk drive is equipped with a actuator locking device to prevent damage to the disks and heads during transportation. Before powering up the drive the actuator must be unlocked. To unlock the actuator, locate the plastic locking knob located on the underside (toward rear) of the drive and turn the knob manually or with a coin from ON (locked) to OFF (unlocked). Check that the cover portion of the latch gears with the concave portion. To lock the actuator reverse the above procedure.

CAUTION

THE UNIT SHOULD NEVER BE SHIPPED OR MOVED WITHOUT LOCKING THE ACTUATOR TO PREVENT DAMAGE TO THE HEADS.

CONTROL AND INDICATORS

Operator control and indicators are located on the operator control panel (front panel) and CNAM and CMZM PCB'S.

The functions of the LED'S and switches on the operator panel are as follows:

Power indicator (Red)	Indicates power is ON.
Ready indicator (Red)	Indicates the initial seek is complete or it indicates the termination of a seek or RTZ operation.
Check indicator (Red)	Indicates a fault condition.
Protect indicator (Red)	Indicates that writing is inhibited.
Protect (PTCT) switch (White)	This switch inhibites the write operation.
Check switch (gray-flat key)	Resets a device Check status.

The fault display LED'S and Mode Select switches, as shown in Fig. 1, are located on the CNAM PCB (Top PCB). Their description is as follows:

FPT indicator (Red)	Indicates that writing is inhibited. (Controlled by SW1-Key 7. ON = In-hibited)
RDY (Ready) indicator (Green)	Indicates that the initial seek is completed or the termination of a seek or RTZ operation has occurred.
ST1 to ST8 (Status 1 to 8) LED'S (Red)	Fifteen fault statuses are visible by binary code as shown in Table 1.
SW1 to SW3	Mode Select switches - These switches must be set as follows to meet the TGSI system requirements:

KEY	SW1	SW2	SW3
7	OFF	ON	OFF
6	ON	ON	OFF
5	OFF	ON	OFF
4	ON	OFF	ON
3	OFF	OFF	OFF
2	OFF	OFF	OFF
1	OFF	ON	OFF

Three LED'S as shown in Fig. 2 are located on the CMZM PCB (Under CNAM PCB). Their description is as follows:

ACDM (Orange)	Indicates that acceleration mode in DC motor control is activated. The LED comes on at power on and stays on until the spindle has reached its nominal speed of 3600 RPM, ±1%, (approx. 20 sec.). At this point the mode changes to inertia and the LED goes out. As the spindle speed decreases to within 1% of 3600 RPM, the mode again changes to acceleration, and the LED comes on.
SPGD (Orange)	Indicates that the spindle speed is within 6% of nominal. This LED will not lite until approx. 20 seconds after power is applied.
PWRDY (Green)	Indicates that +5v, -12V, +24V and the internal + 12V DC are within their nominal range.

POWERING UP/DOWN

Before powering up the drive verify that the actuator has been unlocked, Mode Select switches SW1 to SW3 are set as indicated above, and that the PTCT switch is off (Protect light OFF). The disk drive is now ready to power up.

The disk drive is not equipped with a power ON/OFF switch. Powering up is performed by turning the power switch on the Power Supply to ON (See Power Supply section). After the initial powering up, the drive will normally be powered up/down with the system. Therefore, leave the power switch in the ON state at the end of check-out.

Turn power ON and verify that the spindle comes up to speed within approx. 20 seconds and that the ready light comes ON. Verify that the PWRDY light (On CMZM card) is ON and that the fan is working. The drive is now ready to receive data.

MAINTENANCE

No scheduled maintenance of the disk drive is required. However,if problems should arise, verify that the power supply voltages are correct and that all cables are properly inserted and secure before proceeding.

The sealed module containing the heads, disk, and positioning mechanism is not field serviceable and must be returned to the factory for repair.

Trouble shooting the drive, consist of analyzing all indicators & status LED'S and replacing PCB'S as required. Replacement of the PCB'S, should be based on the basic card function descriptions listed below and the status lights. The PCB'S are easily removed by removing the cover(s), unplugging the connectors, and then removing the screws securing them to the chassis.

The three PCB'S and their basic functions are as follows:

1. The CNAM PCB assembly (top PCB) contains the interface circuitry and is equipped with main-

Table 1. Fault Indicators

Status Bit				Code (Hex)	Fault Status	
ST8	ST4	ST2	ST1		Fault	Description
0	0	0	1	1	DC motor failure (DMFL)	indicates spindle motor failure.
0	0	1	0	2	VCM over heat (VCMHT)	indicates VCM over-heating.
0	0	1	1	3	Initial seek time out (INTMOT)	indicates initial seek has terminated with time-out.
0	1	0	0	4	Control check 1 (CTCK1)	indicates that a Read/Write command was issued during busv status.
0	1	0	1	5	Control check 2 (CTCK2)	indicates that write gate was issued during a fault: condition.
0	1	1	0	6	Read/write check 1 (RWCK1)	indicates that write gate was issued during off.· track.
0	1	1	1	7	Read/write check 2 (RWCK2)	indicates that write current did not flow to the: head during a Write operation.
1	0	0	0	8	Read/write check 3 (RWCK3)	indicates that write gate was issued during File-Protected status.
1	0	0	1	9	Read/write check 4 (RWCK4)	indicates that write gate was issued during a multi-head-selected status.
1	0	1	0	A	Time-out (TMOT)	indicates that seek or RTZ sequence was not: terminated within 500 ms.
1	0	1	1	B	Seek guard band (SEKGB)	indicates that a guard band was detected during a direct seek operation.
1	1	0	0	C	Linear mode guard band (LNMGB)	indicates that a guard band was detected during a linear mode.
1	1	0	1	D	RTZ outer guard band (RTOGB)	indicates that an outer guard band was detected during a RTZ operation.
1	1	1	0	E	Over-shoot check (OVSHT)	indicates that the head over-shot the new cylinder address during settling time.
1	1	1	1	F	Illegal. cylinder check (ILCYL)	indicates that an illegal cylinder address (>588) was issued bv the controller.

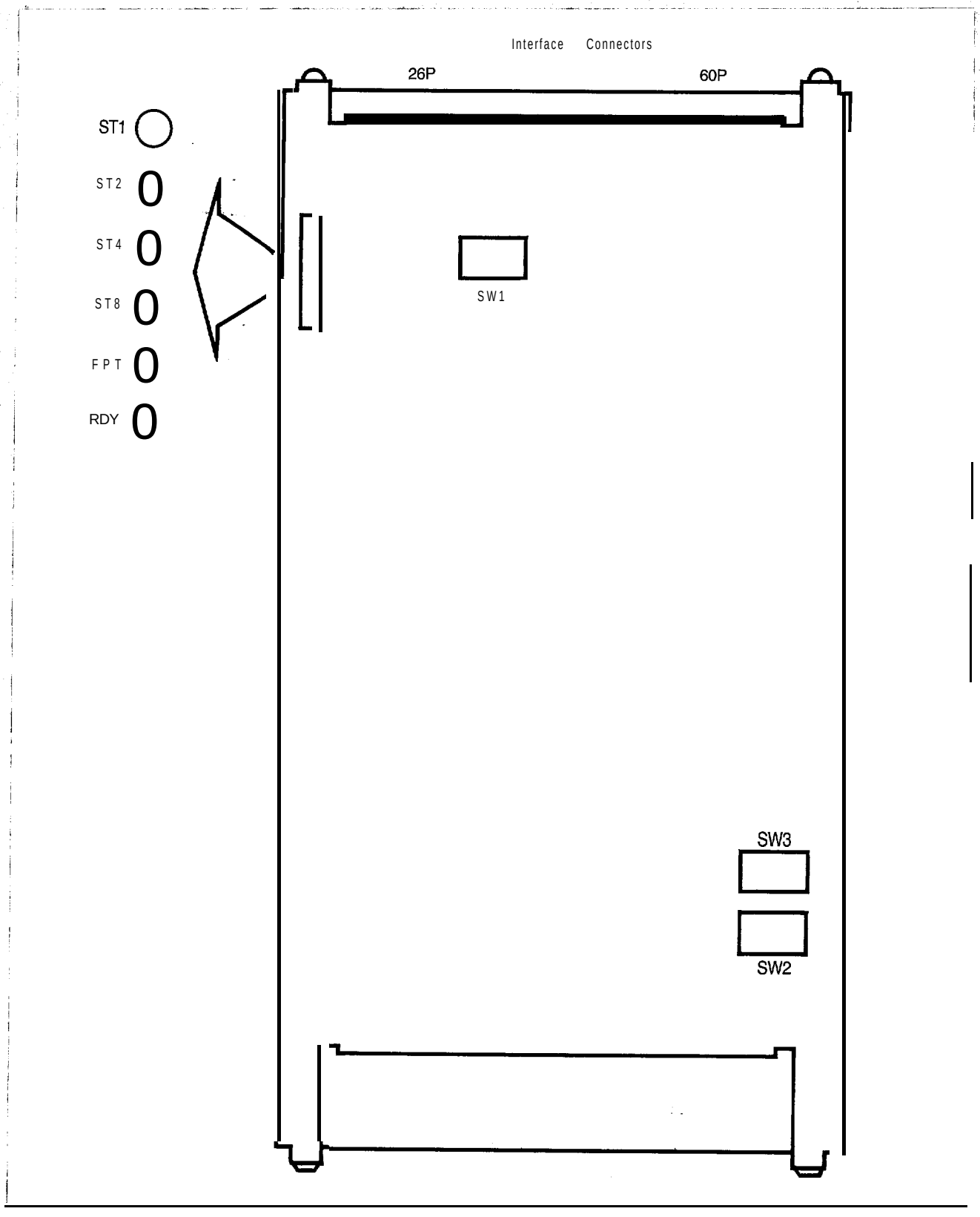


Figure 1. Fault Display Location On CNAM PCB

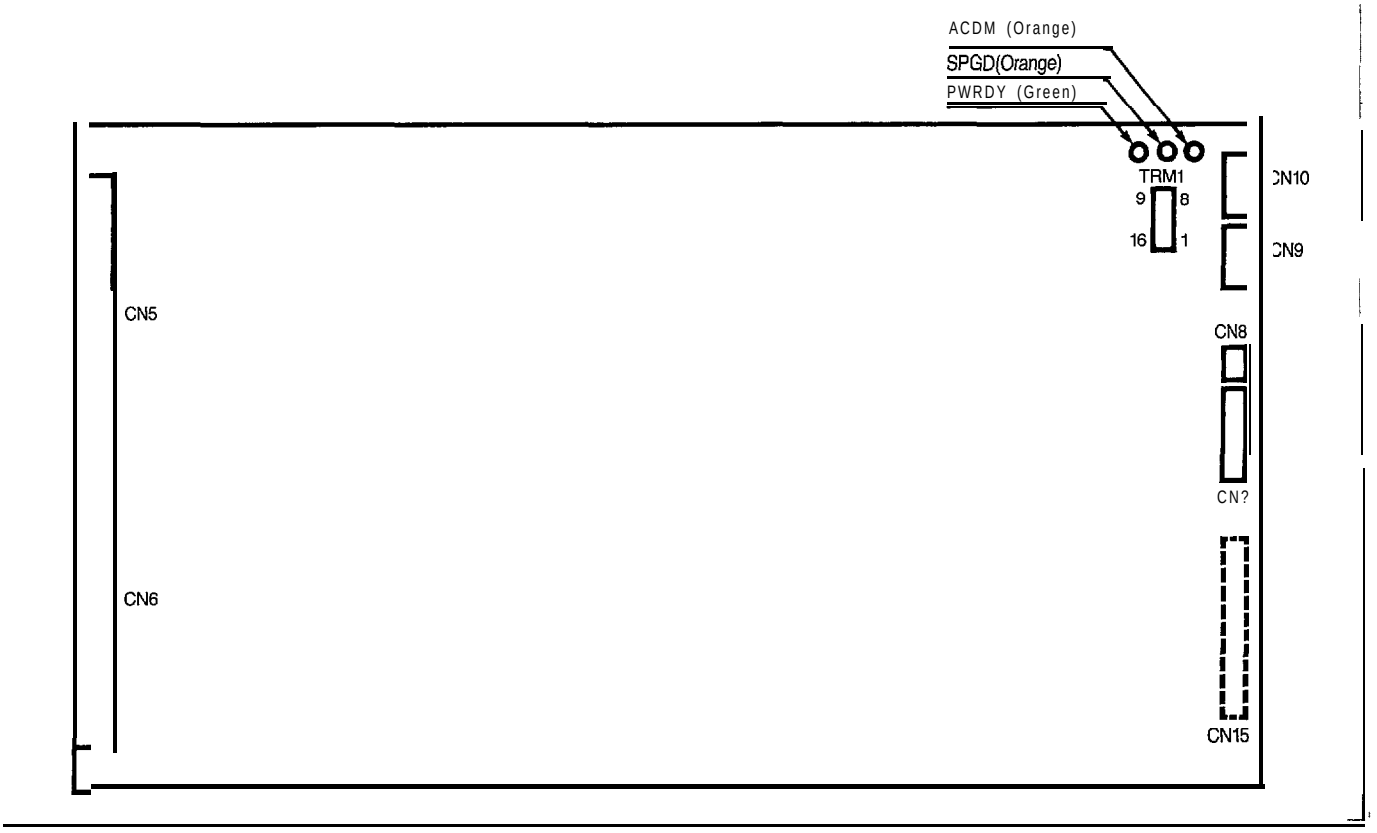


Figure 2. CMZM PCB Assembly

tenance aid LED'S and Mode Select switches- (See Fig.1).

The SMD interface is enabled/disabled by the controller using the drive select function. This card also, makes available **to** the controller the drive's statuses such as On Cylinder, Fault, Seek Complete, Write Protect, Index and Ready.

2. The CMZM PCB assembly, located under the CNAM PCB, contains the read/write and servo control circuits, as well as, additional maintenance aid LED'S (See Fig. 2).

The read/write control includes conversion of NRZ interface write data to MFM coding. When the drive is instructed to read, the head select circuit decodes the selected head and the MFM data *is* converted back to NRZ for transmission to the controller.

The servo control circuits take the cylinder address command from the interface and control the seek operation to any desired cylinder.

3. The TUZM PCB assembly, located behind the cover on back of unit, contains the spindle motor

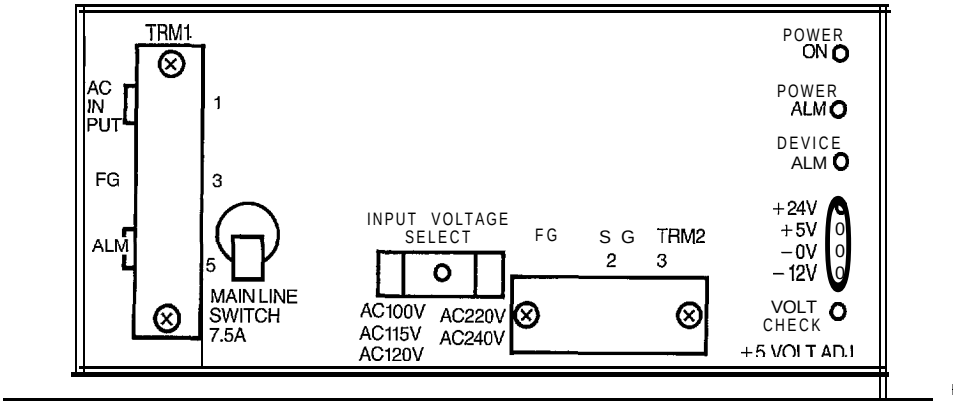


Figure 3. Front View of Power supply

control circuitry which maintains the spindle speed at 3600 RPM ± 1%.

POWER SUPPLY

The M231XK Power Supply (See Fig. 3) capable of operating over a temperature range of O-50" C is provided to supply the required voltages to the disk drive unit as follows:

Output Voltage	Rated Current
+5VDC ± 5% (adjustable ± 10%)	4.8A
+ 24VDC ± 10%(when + 5VDC is nominal)	4.2A
-12VDC ± 5%(when +5VDC is nominal)	4.0A
115VAC + 15%/-24% (For Fan)	0.3A

The unit weights approx. 12 LBS.

The power supply is switch(SW1) selectable to operate from a primary power source of 100/115/120VAC or 220/240VAC.

Input-voltage is applied between TRM1-1 to 2 at the front of the Power Supply. TRM1 pin 3 is provided for frame ground(FG). DC output voltages to the disk drive are provided on connectors CN31 and CN32 at the back of the Power Supply. Connections are as follows:

Connector	No.	Voltage
CN31	1,2	0V
	3,4	-12v
	5,6	0V
	7,8	+5V
	9,10	ov
	11,12	+24V
CN32	1,2	115VAC
	3	FG
	4	
	5	ALM (Receive)
	6	–
CN33(not used this application)	1	+5V
	2	ov
	3	-12V
	4	–
	5	ov

The +5VDC/Adj ± 10% circuit has an over-voltage protection feature which operates if the output voltage exceeds 25% of nominal. The +5V, +24V, and -12 VDC circuits have over current protecton which will reduce the output voltages. The 115VAC circuit for the power supply and disk drive fan is protected by a 250VAC, 1.25A Fuse.

OPERATION

Main Line Switch

This switch controls application of AC power to the power supply. Turning on the switch also applies power to the disk drive and fan. Power-on is indicated by the lighting of the "Power On" LED.

Power Alarm

The power supply is provided with an alarm circuit. This circuit detects improper voltage, excessive temperature rise, drive fan stopped, device alarm, etc. If triggered, the circuit turns off the DC output, lights the "Power Alarm" lamp, and outputs the alarm signal. To clear the alarm condition turn off the main line switch and power up again.

Device Alarm

The Device Alarm indicates that the thermal switch has been closed on the disk drive's fan. The Device Alarm circuitry shuts OFF the voltages to the disk drive to prevent possible damage.

Alarm Signal

The Alarm Signal is a signal output on TRM1-4 and 5. This signal is supplied when either the Power Alarm or Device Alarm circuits are activated and is for customer use only.

MAINTENANCE

The power supply needs no preventive maintenance. If servicing is required, always turn power OFF before starting to service the unit. If unit must be removed for servicing, remount using the same mounting hardware (hardware must not extend into the power supply more than 6mm).

CAUTION

LONGER SCREWS COULD CAUSE AN ELECTRICAL SHORT IN THE POWER SUPPLY.

Replace fuse (AC Output) with same type and size as originally supplied (250V 1.25A).

The fan may be replaced by removing the screws(2) holding the fan and cover in place, then disconnecting

the grounding lead and connector from the fan. Reverse the procedure to install the replacement fan.

**** NOTE ****

Manuals containing more complete information on the Disk brive and Power Supply may be obtained from Fujitsu America Inc., 3055 Orchard Drive, San Jose, Ca. 95134.



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Printed in U.S.A.